Approximate areas under graphs

1

a

i 31.193 units^2 ii 84.791 units^2 iii 51.483 units^2

b $e^4 - 1 \text{ units}^2$

c 4%

2

a

i 8.418 units^2 ii 9.331 units^2 iii 8.928 units^2

b 8.8827 units^2

c 0.51%

3

a

i 1.239 units^2 ii 1.149 units^2 iii 1.016 units^2

b 1 unit^2

c 2%

4

a $f'(x) = e^{-x} - xe^{-x}$

b No solution provided.

c

i 1.034 units^2 ii 0.861 units^2

d $\dfrac{2}{e} - \dfrac{6}{e^5} \text{ units}^2$

e

	2 rectangles	4 rectangles
Percentage error	48.71	23.83

5

a 40.5984 units^2

b 30.9605 units^2

c As $f(x)$ is convex over the interval $1 \le x \le 3$, an underestimate will include a smaller error in the calculation than an overestimate. Hence, the calculation in part (b) will be closer to the exact value.

6

a 8.008 units^2

b 9.594 units^2

c As $f(x)$ has a gradient closer to horizo[redacted]or $2 \leq x \leq 4$ than $0 \leq x \leq 2$, the rectangles used in the overestimate in part (b) are closer to the actual shape of the function. Hence, the calculation in part (b) includes a smaller error in the calculation and will be closer to the exact value.

d As $f(x)$ is concave over the interval $0 \leq x \leq 4$, an overestimate will include a smaller error in the calculation than an underestimate. Hence, an overestimate will be closer to the exact value.

Definite integration

7

a $\frac{16}{3}$ units^2 b $\frac{32}{3}$ units^2 c $1:2$

8 $1:7$

9 $k = \frac{\pi}{6}$

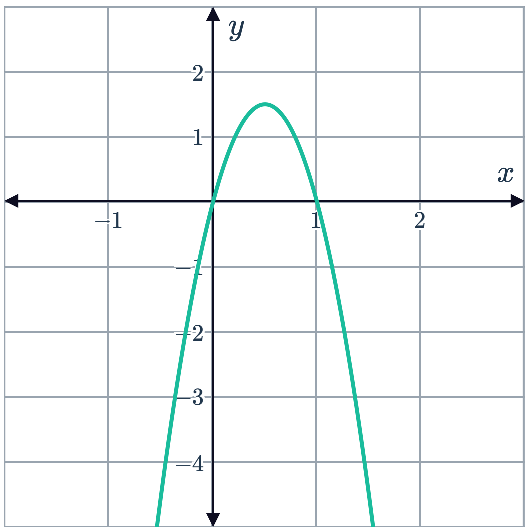
10 $B = A^2$

11

a $7\pi^2 + 2$ b $4\pi^2 + 2$ units^2 c $12\pi^2 - 2$ units^2 d $1:2.81$

12

a



b 1 unit^2

c

a	1	2	3	5	[illegible]
Area $(\text{units})^2$	1	8	27	125	1000

d The area bounded by $f(x) = 6ax - 6x^2$ and the x-axis, for $a \geq 0$ is a^3 units^2.

e No solution provided.

13

a $k = 6$

b $\frac{1}{12}$ units^2

c 12

d

n	1	2	3	4	5
k	6	12	20	30	42

e For the area under the curve of $f(x) = kx^n(1 - x)$ on the interval $[0, 1]$ to be 1 unit^2, $k = (n + 1)(n + 2)$.

f No solution provided.

14

a 1 unit^2

b

i $\cos 2x$ ii $0 \leq x \leq \frac{\pi}{4}$ iii $\frac{1}{2}$ units^2

c $\frac{1}{3}$

d The area bounded by $y = f_k(x)$ and the x-axis over $0 \leq x \leq \frac{\pi}{2k}$, for any positive integer value of k is $\frac{1}{k}$ units^2.

e No solution provided.